ANALYSIS TOOLS

01 CORE ANALYSIS

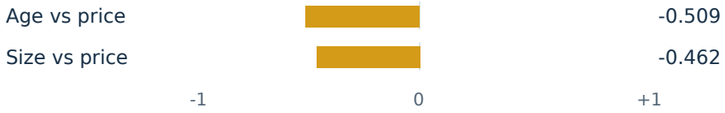
THE SAME 12 EXAMPLE HOUSES

Neighborhood	Sizes (sq ft)	Matching prices ($k)
Central	1000, 1200, 1400, 1600, 1800, 2000	300, 340, 320, 390, 370, 410
Suburban	2200, 2400, 2600, 2800, 3000, 3200	200, 240, 220, 290, 270, 310

1 feature-target-associations

Tests each feature against price separately to find relationships worth exploring.

Ages in each neighborhood: 60, 50, 40, 30, 20, 10



Older houses and larger houses each correlate with lower prices here. Neither result adjusts for neighborhood.

2 target-by-numeric-bins

Groups a numeric feature into ranges, then shows the average target in each range.

3 equal-count size bins; 4 houses per bin



Larger-size bins have lower mean prices because they mix neighborhoods. The correlation still uses all original rows.

3 target-by-group

Compares average price across categories, such as neighborhoods.

12-house calculation

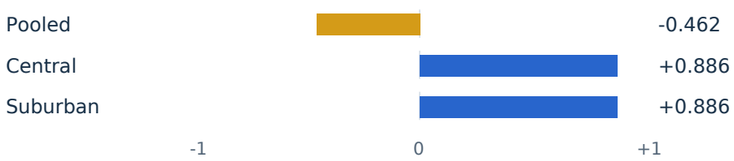


Central averages $100k more than Suburban. This is a group-level price gap, without adjustment for house size.

4 correlation-by-group

Checks the size-price relationship separately inside each neighborhood.

12-house calculation

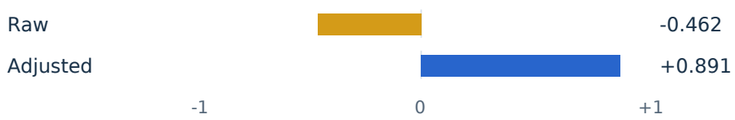


The direction reverses: within both neighborhoods, larger houses cost more. The pooled result mixes different groups.

5 adjusted-correlation

Measures a relationship after removing associations with specified controls.

Control: neighborhood; calculation on ranks



After neighborhood adjustment, size and price correlate positively. This matches tool 4's combined within-group r.

6 nonlinearity-check

Checks whether a curve explains a relationship that a single correlation can miss.

Separate illustration; size (sq ft) vs price ($k)

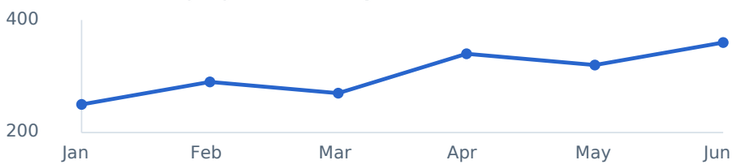


This separate example is U-shaped: middle-size homes are cheaper. Spearman r = 0 misses the curve.

7 time-trend

Groups sales by month and shows how the average price moves over time.

Listed house order: Jan-Jun in each neighborhood



Mean price generally rises (r = +0.886 with time). These are different houses each month, not appreciation of the same house.

8 predictive-feature-importance

Fits features together; shuffles each one to see how much prediction worsens.

Illustrative scores; not a model fitted to 12 houses



Illustration: shuffling size lowers holdout R-squared from 0.80 to 0.50, a 0.30 drop. Importance is not a causal effect.

9 multiple-testing-adjustment

Adjusts p-values across unique tests so a large search is less likely to yield false discoveries.

Illustrative test family; BH adjustment

A	p = .010	q = .030	PASS
B	p = .040	q = .060	no
C	p = .200	q = .200	no

At q < 0.05, only A passes in this example. Repeated references to the same test count once; effect sizes stay unchanged.

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02 PANEL ANALYSIS + FEATURE OVERLAP

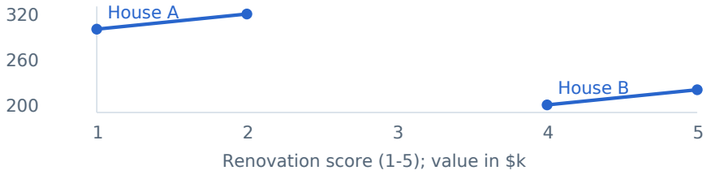
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10 within-between-correlation

Separates one relationship within the same houses over time from differences between houses.

Separate repeated-house illustration; time effects off

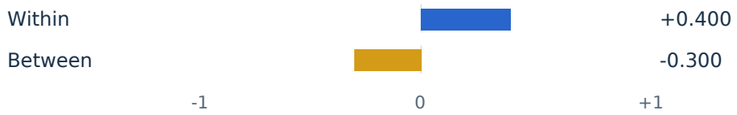


Within A and B, value rises as renovation score rises. Between houses, B has higher renovation scores but lower values.

11 within-between-regression

Estimates within- and between-house slopes while considering several numeric drivers together.

Illustrative renovation slopes; control: condition



Renovation has opposite adjusted slopes within vs between houses. The two models use separate standardizations; these are not correlations.

12 correlation-matrix

Shows which numeric features move together, helping identify overlapping information.

12-house calculation; Spearman r

	Size	Age	Price
Size	1.000	-0.495	-0.462
Age	-0.495	1.000	-0.509
Price	-0.462	-0.509	1.000

Size and age correlate at -0.495: larger homes tend to be newer here. This checks feature overlap, not independent importance.